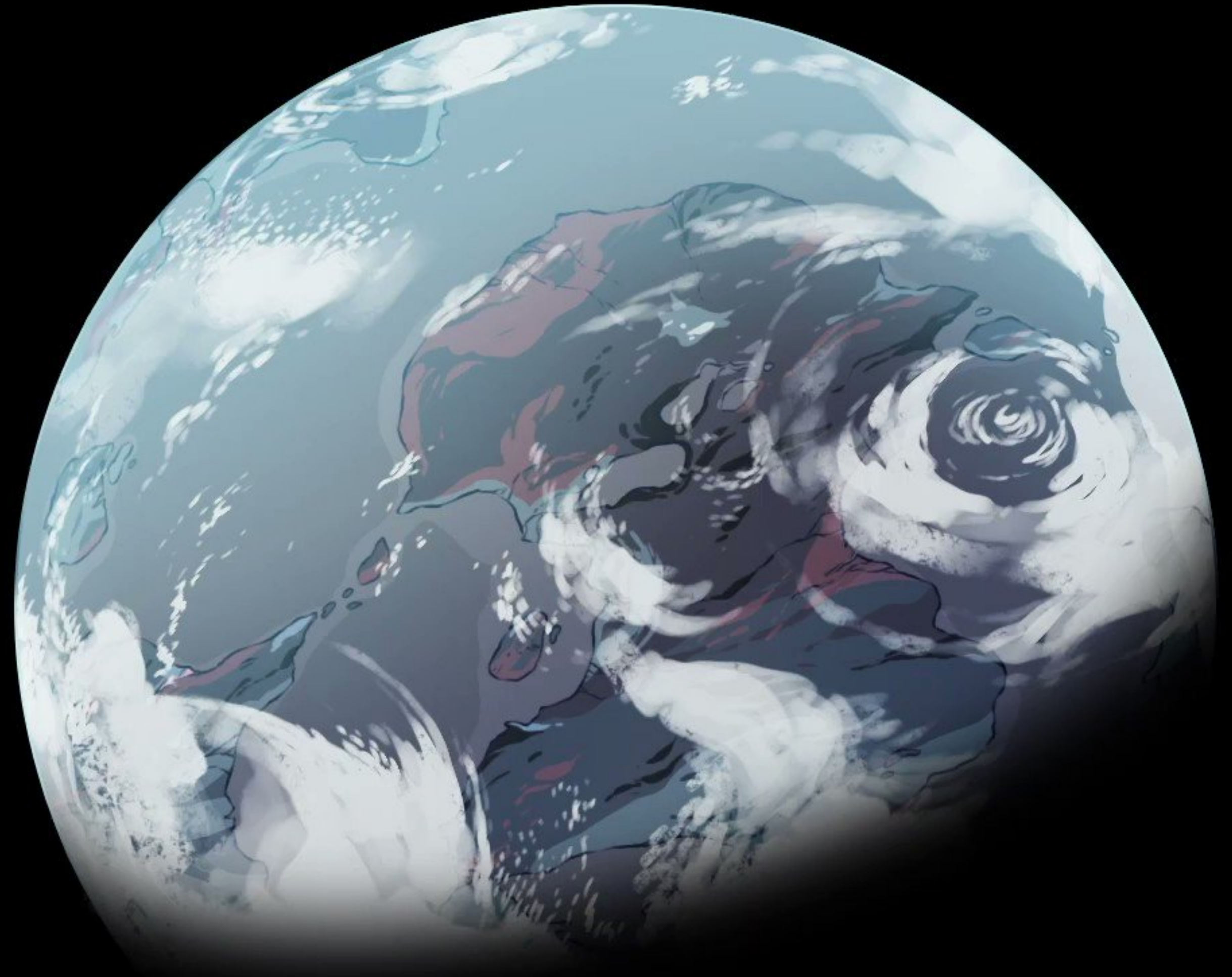


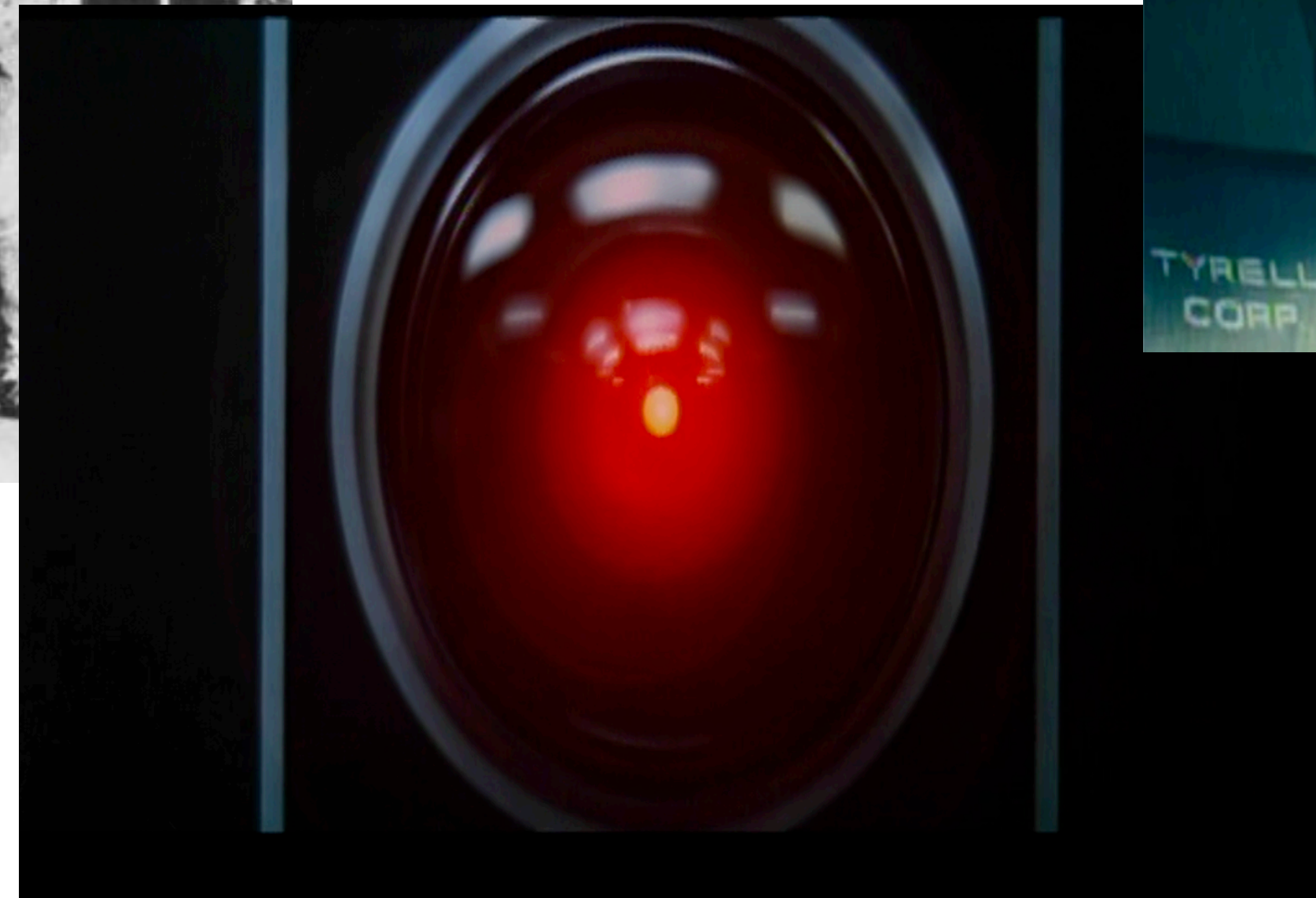
Science & Science Fiction

33-120 | Prof. Katie Breivik

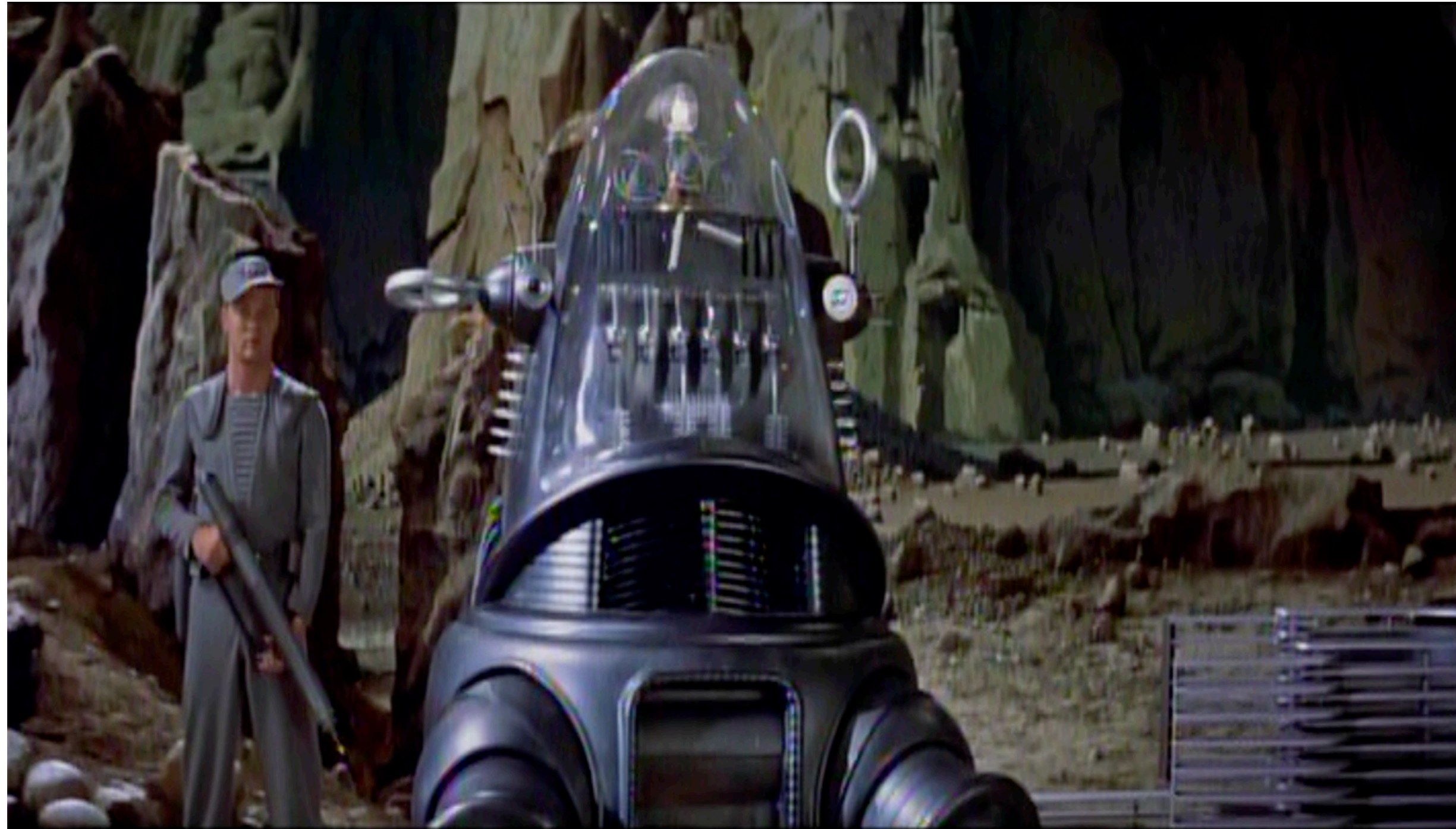


- ▶ Exploration paper 3 is available
 - ▶ due Wednesday, March 19 @ midnight
 - ▶ Can use Chat GPT — see Canvas for details
- ▶ Today: Asimov's Laws of Robotics & Machine Learning

Week 8	03/03/25 No Class: Spring Break	03/05/25 No Class: Spring Break	03/07/25 No Class: Spring Break
Week 9	03/10/25 Intelligent robots	03/12/25 The Turing Test Quiz 3: Matter and Energy (85 pts)	03/14/25 Asimov's Laws and robot behavior
Week 10	03/17/25 Artificial consciousness & robot ethics	03/19/25 Are we alone in the Universe? Conditions necessary for life to exist Exploration Paper 3: AI (25 pts)	03/21/25 Crewed/un-crewed spaceflight Finding habitable planets
Week 11	03/24/25 Finding habitable planets continued Quiz 4: AI (50 pts)	03/26/25 The science behind the SETI project	03/28/25 No class: SETI Team Project Exploration Paper 4: SETI (50 pts)
Week 12	03/31/25 What does it mean to be human? Insight from Frankenstein Medical technology Quiz 5: Extraterrestrial Intelligence (30 pts)	04/02/25 Cell structure and radiation damage DNA, genome sequencing & gene editing SETI Team Project Due (25 pts)	04/04/25 No Class: Spring Carnival
Week 13	04/07/25 Human teleportation	04/09/25 What can we learn from an android about what it means to be human?	04/11/25 How can we solve our problems Public perception of science The scientific method Exploration Paper 5: Being Human (25 pts)
Week 14	04/14/25 Misplaced faith in science Misunderstanding of science Quiz 6: Being Human (50 pts)	04/16/25 Misuse of science in advertising Final Film Critique (50 pts)	04/18/25 What lies ahead? Examples of sci-fi predicting the future How sci-fi storytelling changes with time
Week 15	04/21/25 No class	04/23/25 In class critiques Quiz 7: Problem solving (30 pts)	04/25/25 In class critiques Last day of class – closing discussion



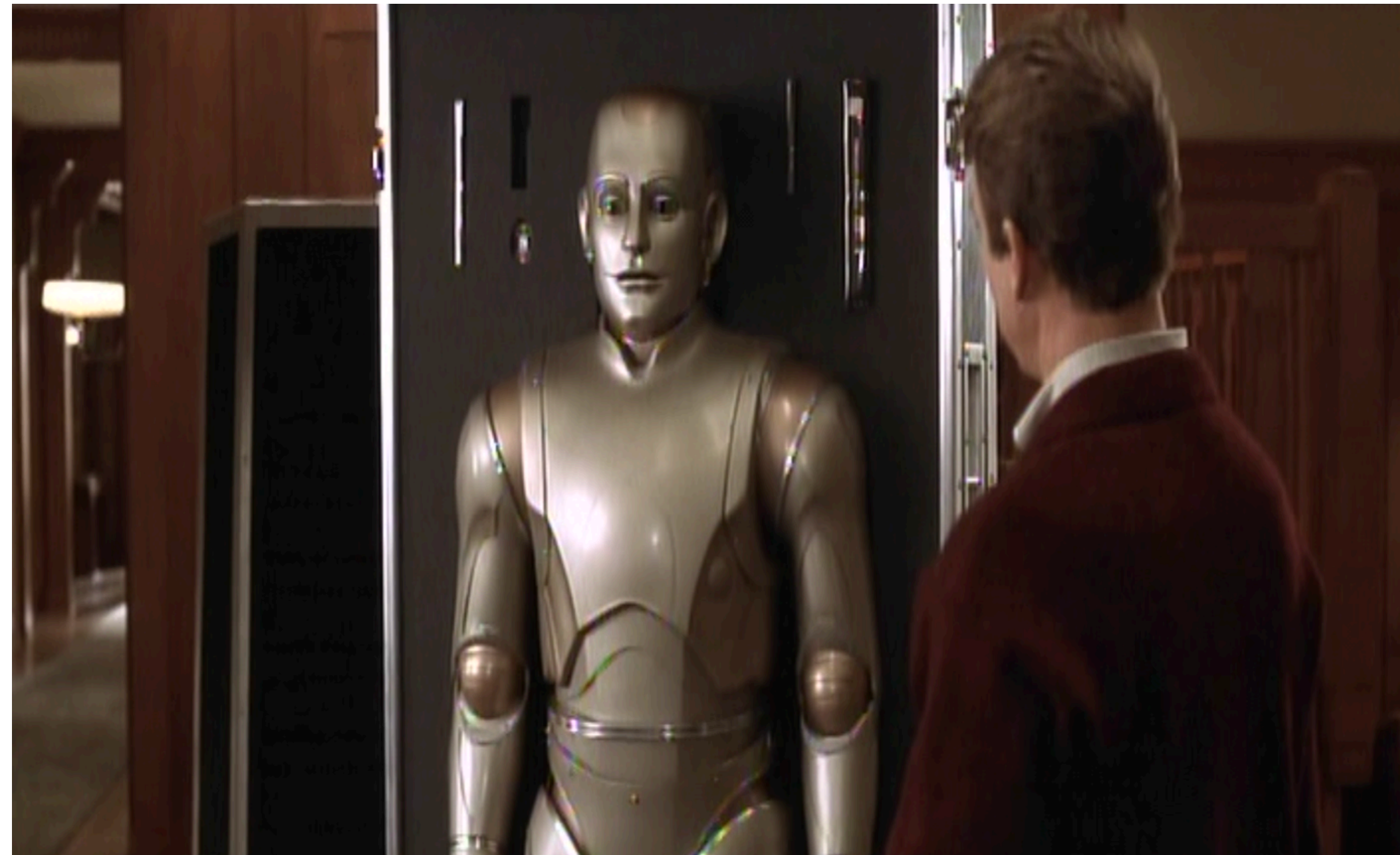
A central theme to intelligent robots in media is the potential danger of such a sophisticated machine



Forbidden Planet

Directed:
Fred McLeod Willcox

MGM (1956)



Bicentennial Man

Directed:
Chris Columbus

Columbia Pictures
(1999)

Asimov's Laws of Robotics:

Isaac Asimov - 1942 - Runaround

"Handbook of Robotics, 56th Edition, 2058 A.D."

- ▶ A robot may not injure a human being or, through inaction, allow a human being to come to harm.
- ▶ A robot must obey orders given it by human beings except where such orders would conflict with the First Law.
- ▶ A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.

WHY ASIMOV PUT THE THREE LAWS OF ROBOTICS IN THE ORDER HE DID:

There
truly
is an
xkcd
for
every
thing

POSSIBLE ORDERING

CONSEQUENCES

1. (1) DON'T HARM HUMANS
2. (2) OBEY ORDERS
3. (3) PROTECT YOURSELF

[SEE ASIMOV'S STORIES]

BALANCED
WORLD

1. (1) DON'T HARM HUMANS
2. (3) PROTECT YOURSELF
3. (2) OBEY ORDERS

EXPLORE
MARS!



HAHA, NO.
IT'S COLD
AND I'D DIE.

FRUSTRATING
WORLD

1. (2) OBEY ORDERS
2. (1) DON'T HARM HUMANS
3. (3) PROTECT YOURSELF



KILLBOT
HELSCAPE

1. (2) OBEY ORDERS
2. (3) PROTECT YOURSELF
3. (1) DON'T HARM HUMANS



KILLBOT
HELSCAPE

1. (3) PROTECT YOURSELF
2. (1) DON'T HARM HUMANS
3. (2) OBEY ORDERS

I'LL MAKE CARS FOR YOU,
BUT TRY TO UNPLUG ME
AND I'LL VAPORIZE YOU.



TERRIFYING
STANDOFF

1. (3) PROTECT YOURSELF
2. (2) OBEY ORDERS
3. (1) DON'T HARM HUMANS



KILLBOT
HELSCAPE



Robot & Frank

Directed:
Jake Schreier

Sony Pictures
(2012)

Is Robot violating Asimov's Laws here?



I, Robot

Directed:
Alex Proyas

20th Century Fox
(2004)

- ▶ Exploration paper 3 is available
 - ▶ due Wednesday, March 19 @ midnight
 - ▶ Can use Chat GPT — see Canvas for details
- ▶ Today: Asimov's Laws of Robotics & Machine Learning